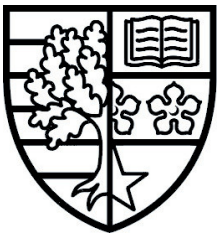
Skin Lesion Classification on FPGA

by Lewis Turner

Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

Supervised by
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Heriot-Watt University
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• Google Gemini 4.0 Argon (thinking: Low - High) to assist setting up FPGA for use over Wifi.

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Acknowledgements

Acknowledgements text.

Abstract

Abstract text.

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1 Introduction

Summary of project motivation & context

1.1 Background and Motivation

- Using papers (mainly) from literature review to provide background context
- Motivations for doing this project
- Identifying problems to be solved to achieve objectives

1.2 Aim and Objectives

State project aims
Table of objectives with IDs to reference later

O1 Train/download a skin lesion model
O2 Compress skin lesion model successfully for FPGA
O3 Run compressed model on FPGA successfully

Testing referring to O - [O3]

2 Literature Review

The proposal for this project was to run a skin lesion classification model on an FPGA. I chose the Xilinx PYNQ-Z1 FPGA board as my target hardware because any FPGA board would have been appropriate for this project and my supervisor Tess Watt had one available by donation to Heriot-Watt University from AMD.
The FPGA board has significant memory and resource constraints [2] so Section 2.4 is to find suitable ML compression methods for FPGAs that best retain baseline model performance and best utilise the FPGA.
ML models are deployed to the FPGA using a pipeline framework, Section 2.3 looks into multiple methods for deploying a model to FPGA with different benefits and supported model types.
The final area of research in Section 2.5 is finding existing skin lesion classification models, both unrelated to FPGAs and designed around running on an FPGA.

2.1 Acknowledgements

Sources for literature/statements
Continuing idea by supervisor Tess Watt in her paper but deployed to FPGA instead of RPi [5]
Survey of recent methods for skin lesion classification between 2017 and 2023 [1]

2.2 Search Methods

I used Google Gemini 3.8 Flash on thinking mode High (October 2026 version) to source the most recent academic papers on a particular topic. I would ask it to create a table consisting of the papers: title, year, summary, DOI or similar URL. This allowed me to leverage the LLM's speed for searching the internet like a more powerful keyword search.
I also used Google Scholar to search for papers by particular keywords such as "skin lesion classification".

2.3 Machine Learning Models on FPGA

Skin lesion classification models are image classifier models. This section describes a few options for pipelines that could be used.

Explain here details about FINN framework: general process, benefits, drawbacks, supported model types

2.4 Compressing Machine Learning Models to FPGA

Papers on useful compression techniques for certain model types

2.5 Skin Lesion Classification Models

using ML has greatly improved early detection of skin lesions...

What is a skin lesion classification model?
Type of Machine Learning model trained to identify lesion areas in images of skin areas

[4, 5].
Choosing an appropriate pre-existing model or training my own will determine the baseline performance I will evaluate against later on.

Top datasets:
Ham10000 - contains 10,015 dermoscopic (medical, of skin area) images categorised for 7 different classes of skin lesion. Used as a standardised benchmark for multi-class skin lesion classification because data is verified to very high standard [1].
ISIC Archive and Challenge - multiple releases of thousands of curated dermoscopic images. Used as a standardised competition benchmarks with split training/validation/test sets and evaluation protocols [1].
PH2 - small set containing 200 dermoscopic images with 3 different classes. Used because includes expert manual verification.

Existing models: Multi-modal approach on an FPGA which combines image features with patient metadata [4]. Uses MobileNetV2 architecture.

Architectures:
Fundamental MobileNet paper with X features [3].

2.6 Further Research Recommendations

Identify gaps, need to repeat any research

3 Methodology

Detailed description of necessary methods, requirements, research questions that will be employed/ needed to complete the project

3.1 Requirements

Quick explanation of functional requirements

FR1 Something about the skin lesion model
FR2 Something about the compression
FR3 Something about the FPGA

Testing referring to FR - [FR1]
Quick explanation of non-function requirements

NR1 Something about running speed
NR2 Something about time taken to compile

Testing referring to NR - [NR2]

3.2 Plan for Execution

Explain how I plan to use experiments to fulfil all the requirements & objectives

3.3 Experiment Planning

How I plan to conduct experiment(s) and what research question does it address/answer (detailed)

4 Feasibility

4.1 Preliminary Work

Summary of work carried out to prepare for this project to validate feasibility.

4.2 Evaluation Strategy

Identify for the project:
- What needs evaluating
- Why it should be evaluated
- How it should be evaluated

5 Conclusions

Short summary of this doc

References

[1] Taye Girma Debelee. 2023. Skin Lesion Classification and Detection Using Machine Learning Techniques: A Systematic Review. *Diagnostics* 13, 19 (2023). doi:10.3390/diagnostics13193147

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[3] Andrew G. Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. 2017. MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. arXiv:1704.04861 [cs.CV] https://arxiv.org/abs/1704.04861

[4] Zhihao Mao, Jin Han, and Mingzhu Shi. 2025. Design and Implementation of a Real-Time Embedded System for Early Skin Cancer Detection. *Journal of Engineering and Science in Medical Diagnostics and Therapy* 9, 2 (12 2025), 021202. arXiv:https://asmedigitalcollection.asme.org/medicaldiagnostics/article-pdf/9/2/021202/7529299/jesmdt-25-1029.pdf doi:10.1115/1.4069441

[5] Tess Watt, Christos Chrysoulas, Peter J. Barclay, Brahim El Boudani, and Grigorios Kalliatakis. 2024. Moving Healthcare AI Support Systems for Visually Detectable Diseases to Constrained Devices. *Applied Sciences* 14, 24 (2024). doi:10.3390/app142411474

6 Project Management

6.1 Gantt Chart

Plan of activities to be carried out in semester 1, then future plan for semester 2

6.2 Analysis of Risks

Including mitigation plans

6.3 PLES

A well-researched consideration of any Professional, Legal, Ethical, and Social Issues pertinent to the project. (e.g. codes of conduct (BCS), codes of practice, standards, computer law, ethical decision making, intellectual property, social aspects, copyright, data protection, and so on).

7 Gen AI Summary

Summarising use of GenAI in project/deliverables